THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A fair coin is tossed once. The probability of getting a head is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{3}$

2. The common difference of the AP $4, 7, 10, 13, \ldots$ is: 1

(a) 5 **(b)** 4 **(c)** 2 **(d)** 3

3. In "If a number is divisible by 10, then its ones digit is 0", the conclusion is: 1

(a) a number is divisible by 10

(b) a number is divisible by 10, then its ones digit is 0

(c) it is not true that its ones digit is 0

(d) its ones digit is 0

4. Over 5 weeks of equal length a cyclist's weekly average distances were 32, 54, 42, 49, 23 km per day. The average over the whole period (km per day) is: 1

(a) 39 **(b)** 42 **(c)** 40 **(d)** 41

5. The divisor algorithm applied to 24 produces a list with how many entries? 1

(a) 9 **(b)** 8 **(c)** 7 **(d)** 10

6. In a parallelogram ABCD, $\angle A = 62^\circ$. Then $\angle D$ is: 1

(a) 138° **(b)** 108° **(c)** 128° **(d)** 118°

7. Is the point $(5, 3)$ a solution of $5x + 7y = 46$? 1

(a) No **(b)** Only for $x = 0$ **(c)** Yes **(d)** Cannot be said

8. The volume of a cube of edge 5 cm is: 1

(a) 125 cm^3 **(b)** 112 cm^3 **(c)** 138 cm^3 **(d)** 152 cm^3

9. The ordinate of the point $(2, -6)$ is: 1

(a) -4 **(b)** -5 **(c)** -6 **(d)** -7

10. The zero of the polynomial $x - 5$ is: 1

(a) 4 **(b)** 7 **(c)** 6 **(d)** 5

11. The decimal expansion of $\frac{3}{4}$ is: 1

(a) 1 **(b)** 0.75 **(c)** 7.5 **(d)** 0.075

12. $(x + 3)(x - 3) =$ 1

(a) $x^2 + 6x - 9$ **(b)** $x^2 - 6x + 9$ **(c)** $x^2 - 9$ **(d)** $x^2 + 9$

13. The perpendicular from the centre of a circle to a chord of length 10 cm bisects it. Each half is (in cm): 1

(a) 7 **(b)** 6 **(c)** 4 **(d)** 5

14. The perimeter of a square of side 9 cm is (cm): 1

(a) 37 **(b)** 35 **(c)** 36 **(d)** 38

15. A bag has 2 red, 3 blue and 5 green balls. The probability of drawing a green ball is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{3}$

16. The next term of the AP $2, 7, 12, 17, \ldots$ is: 1

(a) 23 **(b)** 24 **(c)** 21 **(d)** 22

17. The statement "Every prime number is odd" is: 1

(a) true **(b)** true only for prime numbers
(c) false **(d)** true only for even numbers

18. In a 100% stacked bar of a household's monthly electricity use, appliances take 29% of 300 units. The units used by the appliances are: 1

(a) 87 **(b)** 89 **(c)** 88 **(d)** 86

19. The value of $38 \bmod 12$ (remainder on dividing 38 by 12) is: 1

(a) 2 **(b)** 1 **(c)** 3 **(d)** 4

20. In a parallelogram ABCD, $AB = 7$ cm and $BC = 15$ cm. Its perimeter is: 1

(a) 44 cm **(b)** 48 cm **(c)** 149 cm **(d)** 66 cm

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. A fair die is rolled. Find the probability of getting a prime number. 2

22. Write the first four terms of the sequence $t_n = 3n + 1$. 2

23. Write the converse of "If a number is divisible by 10, then its ones digit is 0". Is it true? 2

24. Find the mean of the numbers 55, 27, 50, 44, 54. 2

25. Add 318 and 427 using the column algorithm. State in which columns a carry is produced. 2

26. Three angles of a quadrilateral are 90°, 120° and 105°. Find the fourth angle. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Write the converse of "If a quadrilateral is a rhombus, then its diagonals are perpendicular". Is the converse true? Justify. 3

28. The average marks of 10 students in a test is 47 and that of 30 students is 31. Find the average marks of all 40 students together. 3

29. Find gcd(10, 30) by listing the divisors of both numbers. 3

30. The sides of a rectangle are 9 cm and 12 cm. Find the length of its diagonal and its perimeter. 3

31. Find two solutions of $4x + 6y = 36$ by putting $x = 0$ and $y = 0$. 3

OR

Find the volume, total surface area and lateral surface area of a cuboid of length 5 cm, breadth 15 cm and height 9 cm. 3

32. Write the quadrant of $(5, 8)$, and the reflections of this point in the x-axis and in the y-axis. 3

OR

Complete the table of values for $y = 3x + 1$ at $x = 0, 1, 2$. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. The mean of 12 observations is 50. A new observation 63 is added. Find the new mean. 5

34. Trace this procedure: total = 0; for i = 1 to 14, if i is even add i to total. Write the values added and the final total. 5

35. The angles of a quadrilateral are in the ratio 1:3:5:3. Find all four angles. 5

36. Write $7x + 6y = 43$ in the form $y = mx + c$ and read off the slope and the y-intercept. 5

OR

Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a cylinder of radius 21 cm and height 5 cm. 5

37. For the point $P(3, 4)$, state its quadrant, its distances from the two axes and from the origin, and its reflection in the origin. 5

OR

For the line $y = 2x + 6$, write the slope, the y-intercept, the x-intercept, and say whether the point $(3, 12)$ lies on it. 5

38. Divide 5 by 11 and write $\frac{5}{11}$ as a repeating decimal. After how many digits does the block repeat? 5

OR

Factorise $x^2 + 5x + 6$. 5

END OF QUESTION PAPER — Total: 80 Marks

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